

W6OTX

W6ARA

PAARA NEWSLETTER
VOLUME 76, NUMBER 7, July 2025

K6OTA

K6YQT

PAARAgraphs



The Official Newsletter of the

Palo Alto Amateur Radio Association, Inc.

The Friendliest Club Around

Celebrating 88 years as an *active* amateur radio club—Since 1937<http://www.paara.org/>

ToADS HF: A Layer Cake of Protocols That Nobody Ordered

Ben Glick, KN6UBF

Why send text over HF the easy way when you could bake a layer cake of protocols no one asked for? ToADS HF (Text Over Audio-Digital-Sideband over HF) is a work-in-progress mode that combines GGWave modulation, RTTY-over-audio, and a CAT-controlled HF setup to reinvent the wheel (with extra layers). In this talk, Ben Glick, KN6UBF explores how this unnecessarily inelegant stack works, why it's fun, and what it says about ham radio's enduring love of solving simple problems the hard way. It's a celebration of signal path absurdity, protocol cosplay, and the kind of engineering hobbyists do for no other reason than because we can.

**This meeting will be a
Hybrid Meeting
Zoom and In Person**

Time: July 11, 2025 07:00 PM Pacific Time
Please check <https://www.paara.org> for
Zoom Details

Upcoming Events

July 11 Aug 1 Sept 5	PAARA General Meeting, 7:00 PM to ~9:30PM Zoom and In Person Meeting.
July 16 Aug 20 Sept 17	Board Meeting, 7:00 PM. Everyone welcome! Zoom Meeting, eMail President for details.



President's Corner

I hope everyone's summer is off to a rousing start! This month's PAARAgaphs is arriving a little earlier than usual, especially because the next club meeting is going to be on July 11th, which is the second Friday of July, rather than the 4th of July. I encourage everyone to enjoy the celebrations on the fourth including parades, BBQs, concerts, fireworks or whatever other events are happening near you. I'd also like to especially

*(President — Continued on page 4)*

Question
of the
MONTH

1. How many U-Haul trucks did PAARA use at Field Day?
2. What color was the GOTA (Get On The Air) Canopy?
3. What was served for dinner on Saturday evening?
4. What was flying from the tower trailer at Field Day?
5. What was PAARA's Field Day class and section?
6. Name one (or all) of the station callsigns PAARA used for Field Day?

Soggy SOTA – April 2025 on Black Mountain (W6/NC-150)

**Connie W6EFI, with
Justin AI6YM, Sorin KN6YUH**

(Much of this adapted from the PAARA Discord #sota channel)

Justin AI6YM invited folks along for a Summits on the Air (SOTA) activation at the end of April on Black Mountain, in the Monte Bello Open Space Preserve up Page Mill in Palo Alto. Connie W6EFI and Sorin KN6YUH joined him. To “activate” a peak means making at least 4 contacts while located close to the summit.

We met up at around 10am in the Montebello OSP parking lot. (Parking is free. Sometimes the lot fills up especially on nice weekends but there is additional parking across the street.)

On the drive up beforehand, we coordinated on N6NFI and planned if necessary to switch to the simplex “adventure frequency” but never needed that since we never separated. We loaded up our equipment, changed our shoes, had the last sip of coffee, and hit the trail at around 10:30 am.



Figure 1: The SOTA party sets out

The hike takes you initially down from the parking lot to a low elevation of about 1800' in a canyon along the San Andreas fault. You walk literally along the faultline for a bit followed by a steady uphill hike up to the summit of Black Mountain at about 2800'. The total one way is about 2 1/2 miles. The climb up from the canyon to the summit covers about a mile and a half.

A bit of drizzle emerged when we reached the

backpack camp area shortly before we got to the top. Connie and Sorin tried to wait it out under a tree before we hiked on to the summit.

The summit area we chose for operating was relatively flat and grassy. There was a great view to the north-east of the valley.

Unfortunately, the rain came back as we finished setting up and we had a very wet activation. But we'd already hauled our gear up the d*** mountain. We were determined to get our points.

Sorin and Justin activated on 2m FM with Yagi-Uda antennas -- Justin's on a tripod and Sorin's handheld. Connie activated



Figure 2: No Justin, you can't activate from a pay-phone (at the backpack camp)



Figure 3: AI6YM with his Yagi-Uda

ed 20m CW with 10W and a 17' ground-mounted vertical. Sorin's longest contact was about 60mi with 5W to Mt. Vaca. Connie's farthest QSO that day was Colorado.

Sorin observed some interesting effects of the terrain. He had brought a directional antenna with a plan to carve up



Figure 6: W6EFI activating on 20m CW

(SOTA — Continued on page 3)

the area into different slices that could be worked separately; but he was surprised to talk to someone in Palo Alto while pointing his antenna down towards Cupertino / San Jose. He concluded that this was the result of the signal reflecting from the southeast bay mountains so the “slicing” was not so precise.

them get oriented on the map. This same group caught up to us later on the way down and let us know they had figured it out.

We all took away some notes for ourselves for improvements the next time out. Connie wants to pack her pack more effectively for operating directly out of the pack while carrying first aid and lunch and maybe a chair. Sorin wants to bring along an antenna support so that hands are free while operating with a directional antenna. Justin is always tinkering and testing his inventions as he did on this trip.

73,
Connie W6EFI



Preparing an Antenna for Field Day, what a contrast to SOTA Antennas

(President — Continued from page 1)

acknowledge our club members who are veterans for their courage, dedication, and special commitment and service to our country. Have a great 4th of July!

Hopefully you receive this newsletter just in time for ARRL Field Day. We will begin setting up at Bedwell Bayfront Park in Menlo Park at 5pm on Thursday, June 26th with the erection of one of the full-sized beam antennas. We'll assemble and erect two more antennas starting at 10am on Friday, June 27th. Saturday morning, work will start at 9am to finish setting up the stations, including connecting all the logging computers to the network. The operating period starts at 11am and continues for 24 hours until 11am on Sunday. According to a recent ARRL Member Bulletin, this summer is the peak of Solar Cycle 25, and the sun is very active, which could mean tremendous propagation and a legendary Field Day!

We would love for you to join us for as much of the Field Day activities as your schedule allows. Help with setup and tear down is very appreciated, but please also come by to socialize, get some seat time making contacts for fun and to help the club earn Field Day points, and maybe to try an operating mode or two with which you might not be very familiar.

Field Day has a lot of goals and activities, but one of them is public outreach. PAARA can earn bonus points for things like youth participation (age 18 or younger), site visits by elected officials, and media publicity. So, if you happen to be 18, an elected official, or the owner of a newspaper or TV station (or you know someone who is), come by and check out Field Day with us!

If you have any questions about Field Day, I encourage you to visit the Field Day section of the PAARA web site at <http://paara.org/fieldday.html>. In particular, please make note of the park rules and regulations, and the safety guidelines. We want everyone to be safe, and to be considerate visitors to the park so that we can continue to operate in this location for many years to come.

Lastly, for next month's PAARAGraphs, we'd like to put together a kind of scrapbook covering many different activities and experiences from

Field Day. We will have an area at the site where people can leave notes describing their impressions or something they learned or thought was fun. But we'd also like to encourage members to submit photos with short captions so that we get a good cross section of different people's impressions of Field Day. Look for instructions at the site for how you can submit photos and captions, or you can always just send them directly to me at president@paara.org.

Whether you're able to come out to Field Day or not, I hope to see you at the next meeting on July 11th, either IRL or on Zoom! If you're there in person, please join us after the meeting for pizza and lively conversation at Mountain Mike's across the parking lot. We'd love for you to join us!

73, Bob KN6YGN

Get on the air to keep the airwaves alive!

June 2025 Board Meeting Minutes

In attendance were President Bob Ridenour KN6YGN, Vice President Rob Fenn KC6TYD, Secretary Ric Hulett N6AJS, Treasurer Margaret Cooper K6WEK, Directors Sorin Constantinescu KN6YUH, Doug Teter KG6LWE, and Darryl Presley KI6LDM. Also attending were Connie Stillinger W6EFI, Clark Martin KK6ISP, and Shrikumar H. KA6Q.

The meeting was called to order at 7:05 p.m. A quorum was present.

Officer Reports

- President's Report (Bob):
 - Note that the July club meeting is scheduled for 7/11 because of the July 4th holiday.
 - Domain registration for paara.net was renewed on 6/16.
 - Forms to renew K6OTA and assign new trustee were submitted and expected to be processed to the FCC on 6/16.
 - My apologies for getting PAARAGraphs

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(Continued from page 4)

out a few days late last month. I was confused about when they would be ready and didn't see the notification email come into an unexpected account.

- Secretary's Report (Ric):

- The payment for K6OTA license renewal is not complete, but will be within a day.
- Membership update: We have about 5 new and renewal memberships to process (I've been out of town since last Month).

- Treasurer's Report (Margaret):

The board voted to establish a pro tem committee to explore our bookkeeping needs and options.

- VP / Program Chair Report (Rob):

- Our speaker for the July meeting will be Ben Glick KN6UBF. His topic: "ToADS HF: A Layer Cake of Protocols That Nobody Ordered"

Old Business

Members: Please search your memory banks for interesting Ham Radio stories: Historical, Anecdotal, or Amusing! We continue to need articles for PAARAgaphs. This is your turn – write something up, and send it to our editor Bob KN6YGN.

We have a story for July, thanks to W6EFI

For August, we will cover our Field Day activities: Please contribute your FD photos and stories.

- Education Committee [Darryl]

- Question of the month: For July, we will have several questions related to our Field Day activity, such as: "What flag is on the B station Tower?" and, "What call sign(s) did PAARA use at FD?"
- The board discussed planning for the September PAARA in the Park Activity. This gathering will be September 13 at Rengstorff Park in Mountain View.

Our educational activity will be a soldering project, coordinated by W6EFI and KK6ISP. Connie showed an example of the finished project. The cost of the pro-

ject for participants will be between \$15 and \$25 depending on final material cost.

- Field Day:

- Antenna parties are done, with plenty of pictures up on the website under Photos or Field Day. The Tower and Antenna installation were smooth.
- We will have 4 generators available.
- U-Haul trucks will be picked up at 4:00 on Friday
- Network Day: This will be held Saturday 6/21 at W6NEV, 10:00 AM. Station captains, radios and computers will all be there and set up.
- T-Shirts were distributed at the June club meeting; If not picked up, they will be available for pickup at the FD site.

- The next PAARA Field Trip is to the USS Hornet, on August 23.

- IT Subcommittee Updates:

- We have additional web maintenance volunteers, who will help manage website changes. Ben KN6UBF, Alan AK6AD, Tae N6TAE.

- Coffee Duty [Sorin]

- Bob has pre-measured coffee into vacuum sealed pouches to make it easier to get the coffee set up. Should also be easier to keep the equipment bin clean.

- 900 MHz repeater status: [Christopher]

- We need to replace a network patch cable as rats have chewed on the existing one. A 5' stainless steel armored Cat 6a jumper is ~\$20 on Amazon.

- New Ham Outreach: (Rob) Postcards (63) will be going out this week.

New Business (None)

The meeting was adjourned at 9:05 pm

Respectfully submitted

Ric Hulett N6AJS
PAARA Secretary



June Raffle Winners (left to right):
 Mike (N6WAI), Bob (KN6YGN), Doug (KG6LWE), Joe (KR1JOE), Perry (KI6OPZ), Dean (KJ6IWI)

1st YEASU FT-4XR
 UHF/VHF



PAARA



2nd iFixit Mako Driver Kit



3rd Nitecore NU25 400 UL Ultra
 Lightweight Headlamp



4th Goldstar (LG) Digital Multimeter DM-311

5th VCELINK Wire Stripper + KLEIN TOOLS flush cutter



Barista Opportunity

The club needs your help with setting up the coffee urns before the meeting starts, and with taking them down when the meeting is over. Simple and clear instructions will be available, as well as coaching as needed. If you can also store the two boxes between meetings and wash the baskets from the urns between meetings (particularly before a meeting where you can help with setup) that would be of great help, although it's not strictly necessary. Please add your name to the coffee rotation sign-up sheet which will be posted above the coffee urns, or get in touch with Sorin KN6YUH. Your help is greatly appreciated!

Discord Server

Do you want to talk to other PAARA members and friends about Homebrew, POTA, SOTA, digital modes, or any other ham radio topic? Download the App or use the Web version, then join our server by scanning the QR code, or visiting <https://discord.gg/qPyn5h8p6H>.



Palo Alto Amateur Radio Association, Inc.

PO Box 911 Menlo Park, CA 94026

Officers

President	Bob Ridenour, KN6YGN 650-575-4528 bob_ridenour@yahoo.com
Vice President	Rob Fenn, KC6TYD 650-888-9060 kc6tyd@gmail.com
Secretary	Ric Hulett, N6AJS 408-332-4593 n6ajs@arrl.net
Treasurer	Margaret Cooper, K6WEK k6wek@arrl.net

Directors

Director ('24-'25)	Sorin Constantinescu, KN6YUH, consta@gmail.com
Director ('25-'26)	Walt Gyger, K6WGY 408-921-5901 wgyger@ix.netcom.com
Director ('25)	Doug Teter, KG6LWE 650-743-7892 dteter@wcwi.com
Director ('25)	Darryl Presley, KI6LDM 650 255-2454 ki6ldm@arrl.net

Appointed Positions

Membership	Ric Hulett, N6AJS 408-332-4593 N6AJS@arrl.net
Database	Ric Hulett, N6AJS 408-332-4593 N6AJS@arrl.net
Station Trustee	K6OTA Ron Chester, W6AZ
Property Manager	Doug Teter, KG6LWE
Badge Coordinator	Doug Teter, KG6LWE 650-743-7892 dteter@wcwi.com
Historian Position	<i>Position Vacant</i>
Raffle Coordinators	Rob Fenn, KC6TYD, kc6tyd@gmail.com Shrikumar, KA6Q shri.paara@enablery.org
Field Day Coordinator	Doug Teter, KG6LWE 650-743-7892
ASVARO Rep	Clark Martin, KK6ISP kk6isp@sonic.net
Webmaster	Shrikumar, KA6Q webaron@gmail.com
Technical Coordinator	Christopher, AI6KG 408-348-0304 ch@murgatroid.com
QSL Manager	Ric Hulett, N6AJS 408-332-4593
Speaker Coordinator	Rob Fenn, kc6tyd 650-888-9060

PAARAgaphs Staff

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Member Profiles	<i>Position Vacant</i>
Technical Tips	Ric Hulett, N6AJS
Photographer	<i>Position Vacant</i>

VE Exams

De Anza Park, Sunnyvale, 2nd Saturday 10:30 am each month except November and December. See website for details and exceptions: <http://amateur-radio.org>

Electronics Flea Market (EFM)

Sponsorship: Association of Silicon Valley Amateur Radio Organizations (ASVARO). The Electronics Flea Market is held at West Valley College, 14000 Fruitvale Ave, Saratoga.

Website: <http://www.electronicfleamarket.com/>

PAARA — Palo Alto Amateur Radio Association

Meets 1st Friday 7:00pm each month at Room H-6, Cubberley Community Center; Net 145.230 - PL 100Hz Mondays at 8:30. See website at <http://www.paara.org>. For more information. contact: Joel Wilhite KD6W, kd6w@arrl.net, 650-325-8239

FARS — Foothills Amateur Radio Society

Meets 4th Friday each month at 7:00pm at Covington School, Los Altos.

Website: <http://www.fars.k6ya.org>

NCDXC — Northern California DX Club

Meets 3rd Thursday 7:00pm each month, Repeater for member info 147.360. Contact president@ncdxc.org. Website: <http://ncdxc.org>. YouTube content: "The Northern California DX Club Official Channel". Cohost of the International DX Convention.

The 50MHz & Up Group of Northern California

This organization specializes in vhf + wak signal and microwave activities. Meetings are held on the first Tuesday of each month. Time is usually 5pm for in person meetings, and 7pm for Zoom only meetings. In person meetings are held Sports Basement, 1177 Kern Ave, Sunnyvale. Always check the website, <http://50MhzandUp.org>, for correct information. Zoom information is also there.

San Mateo Radio Club W6UQ.ORG

Meets, 3rd Friday, January through November. Tuesdays & Thursdays, [Directed] Net, 7pm, N6ZX 145.370Hz, -600KHz, PL107.2Hz
Contact: SanMateoRadioClub@gmail.com, Website: <http://W6UQ.org/calendar>

SPECS

Southern Peninsula Emergency Communication System users Group

Meets each Monday 7:30pm and 8:00pm.
See: <https://specsnet.org/monday-night-net> for more info.
Contact: <https://www.specsnet.org/contact> or board@specsnet.org

SCARES

South County Amateur Radio Emergency Service

Meets 3rd Thursday 7:30pm each month, Belmont EOC, Belmont City Hall, One Twin Pines Lane, Belmont CA 94002. Net is on 146.445 [PL 114.8] & 444.50 (PL-100) 7:30 Monday evenings. Contact: President Gary D. Aden, K6GDA 650-743-1265 (D), 650- 595-5590 (N)
Web: <http://k6mpn.org> E-mail: pres@k6mpn.org

SCCARA

Santa Clara County Amateur Radio Association

Operates W6UU & W6UU/R, repeater 146.985-pl
Nets: 2m, 7:30pm Mon; 70cm, 10M (28.385) 8PM Thur.
Meets 2nd Mon each month @ 7:30 PM.
ARRL/VEC license testing contact 408-507-4698

SVECS — Silicon Valley Emergency Communications

Operates AA6BT repeater (146.115 MHz+)
Website: <http://www.svecs.net> or contact: Lou Stierer WA6QYS 408 241 7999

WVARA — West Valley Amateur Radio Association

W6PIY six-meter repeater on 52.58MHz. Normally, six-meters is linked with 147 and 223, while 441 and 1286 repeaters are linked.

VHF: 52.58 (-500) 151.4 ctcss UHF: 147.39 (+600) 151.4 ctcss 441.35 (+5.0) 88.5 ctcss
223.96 (+1.6) 156.7 ctcss 1286.20 (-12m) 100.0 ctcss

Meetings are 2nd Wednesday of every month except July, August and December.

Website: <http://wvara.org>, Contact: info@wvara.org

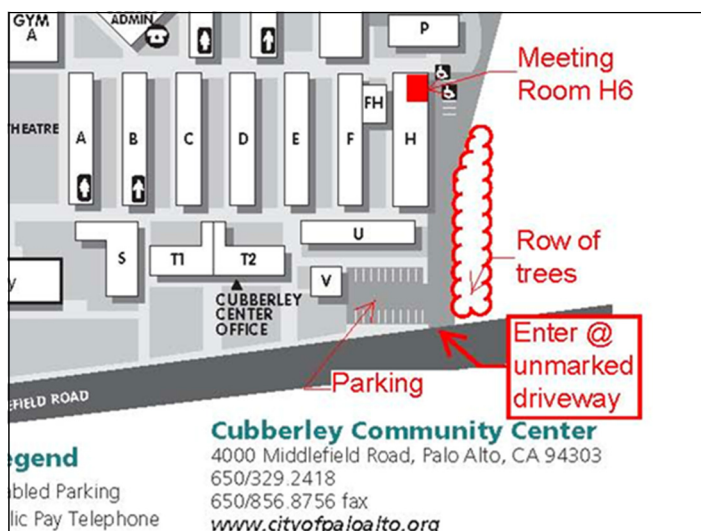
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For Sale: Heathkit Station:

- SB-102 Transceiver/Mike- \$250
- SB-600 Speaker/Power supply - \$50
- SB-220 Amplifier - \$750
- SB-610 Station Monitor - \$50
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PAARA Weekly Radio Net

Info and Swap Session
 every Monday evening at 8:30pm
 on the N6NFI 145.230 MHz repeater

Week Control Operator

1 st	Doug - KG6LWE
2 nd	Doug - KG6LWE
3 rd	Ric - N6AJS
4 th	Rob - KC6TYD
5 th	Rob - KC6TYD

If you're interested in trying out at Net Control, Contact Doug, KG6LWE. It's good practice, and lots o' fun! Give it a try.

Palo Alto Amateur Radio Association P.O. Box 911, Menlo Park California 94026-0911

Club meetings are on the first Friday of each month,
 7:00pm at the Room H-6, Cubberley Community Center.

Radio NET & Swap Session every Monday evening, at
 8:30pm, on the 145.230 –600 MHz repeater, PL 100Hz.

Membership in PAARA is \$25.00 per calendar year,
 which includes one subscription to PAARAgraphs
 \$6 for each additional family member (no newsletter).

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Scroll to the bottom of the page
 and fill out the info. All badges will
 be mailed.

The cost for a badge is \$30.00.

PAARAgaphs Ad Rates

PAARAgaphs accepts paid advertisements from non-members. (short personal ads remain free for members in good standing). **All ad rates listed are per issue.**

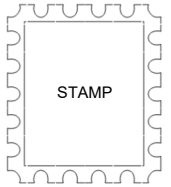
- Not-for-profit ads by association members for ham-related items and wants. No cost for business card-size ads (additional space at \$2.50 per business card size per issue).
- For Profit organizations and/or individuals: \$5-business card size, \$14.00-quarter page, \$25-half page, \$50 full page or back cover per issue.

These fees may be reduced or waived in exchange for a valuable consideration that is given to the Association or its general membership. Such consideration must be in addition to any existing arrangements with the association. The PAARAgaphs editors reserve the right to reject any ad deemed to be not in the best interest of the Association.

All fees payable in advance by the year with "scanner-ready" copy or text-only ads. **Give payment and copy to Walt Gyger, K6WGY.**

PAARAgaphs — July 2025

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PAARAgaphs Newsletter
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- Over 1000 Memory Channels • Wide Bank Receiver Coverage • High Power Output 5 Watts • High/Low Drilling • NOAA Severe Weather Alert with Alert Scan



FT-70DR CFM/FM 144/430MHz Xciv

- System Fusion Compatible • Large Front Speaker delivers 700 mW of Loud Audio Output • Automatic Mode Select: detects CATM or FM Audio and Switches Accordingly • Hange 1, 105 Channel Memory Capacity • External DC Jack for DC Supply and Battery Charging



FT-5DR CFM/FM 144/430 MHz Dual Band

- High-Res Full-Color Touch Screen TFT LCD Display • Easy Hands-Free Operation w/Built-in Bluetooth® Unit • Built-in High Precision GPS Antenna • 1200/800bps AFSK Data Communications • Supports Simultaneous CFM Digital Micro SD Card Slot



FT-65R | 144/430 MHz Transceiver

- Compact Commercial Grade Rugged Design • Large Front Speaker Delivers 1W of Powerful Clear Audio • 5 Watts of Portable RF Power With-in a compact Body • 3.5-Hour Rapid Charge Included • Large White LED Flashlight, Alarm and Quick Home Channel Access



VX-6R | VHF/UHF Dual Band Hand-Held

- Big-Radio Features in a Compact Package • Wide-Band Receiver Coverage for Catching All the Action • 900-Channel 24-Bank Memory System • Outdoor-Ready Features including Waterproof Rating • Smart Screen Automatic 51-Channel Scanning/Locking System



• RETAIL LOCATIONS - Store hours 10:00AM - 5:30PM - Closed Sunday
• PHONE - Toll-free phone hours 8:30AM - 5:30PM
• ONLINE - WWW.HAMRADIO.COM
• FAX - All store locations
• MAIL - All store locations

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